Delta Physics

Our Dynamic Universe
Collisions, explosions, and impulse
5 minutes – Individual

Gravitation

Why does the Moon pull on the Earth?



Higher Physics

PART 2

Newton's Law of Universal Gravitation

Everything pulls on everything else. Why don't you feel the pull of the person beside you?

By the end of this part…

1. I can **use $F = Gm_1m_2/r^2$** to find **force**, **mass** or **distance**
2. I can **find r** from the **centre** of a planet: **$r = R + h$**
3. I can **state** and **justify** how **F** changes with **mass** and **distance**
4. I can **calculate g** at a **satellite**, using **$W = mg$**

Newton's Law of Universal Gravitation

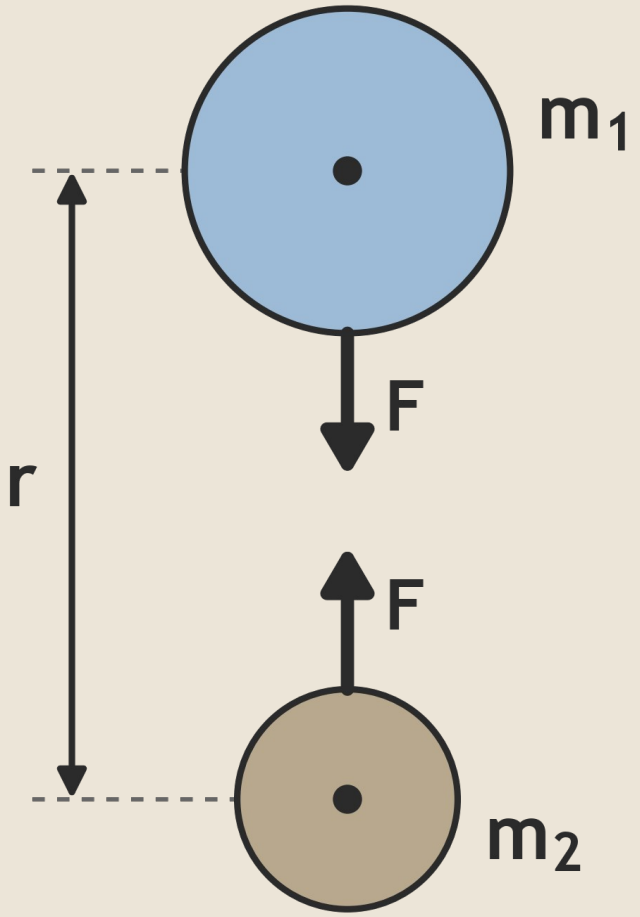
$$F = G\,(m_1\,m_2)/r^2$$

F gravitational force (N)

G 6.67×10^{-11} m^3 kg^{-1} s^{-2} (data sheet)

m_1, m_2 the two masses (kg)

r distance, centre to centre (m)



r: centre to centre

r Is Centre to Centre

r

centre to **centre**

R

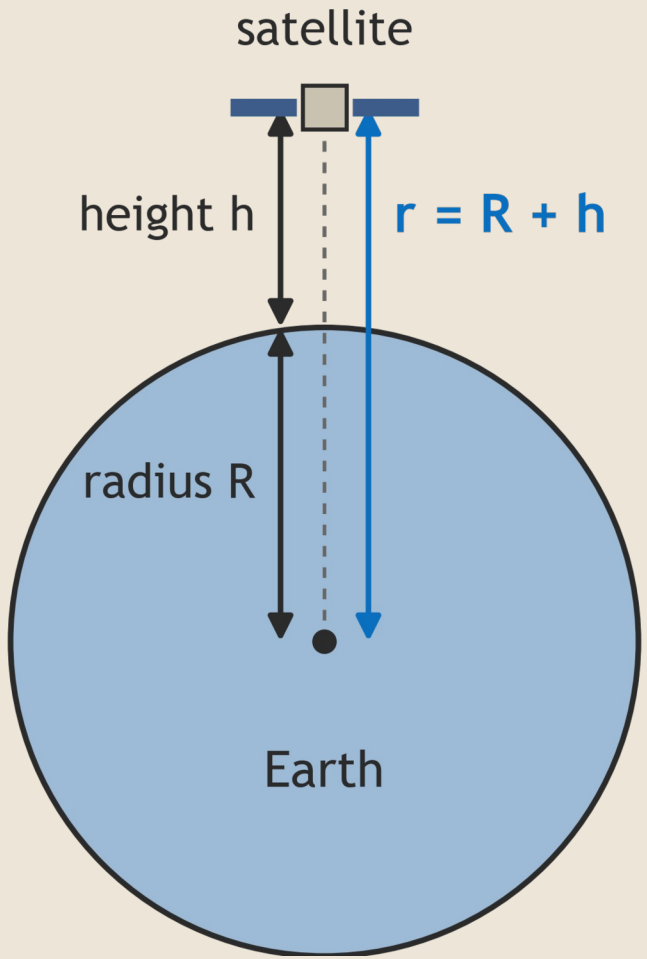
radius of the planet

h

height above the surface

So

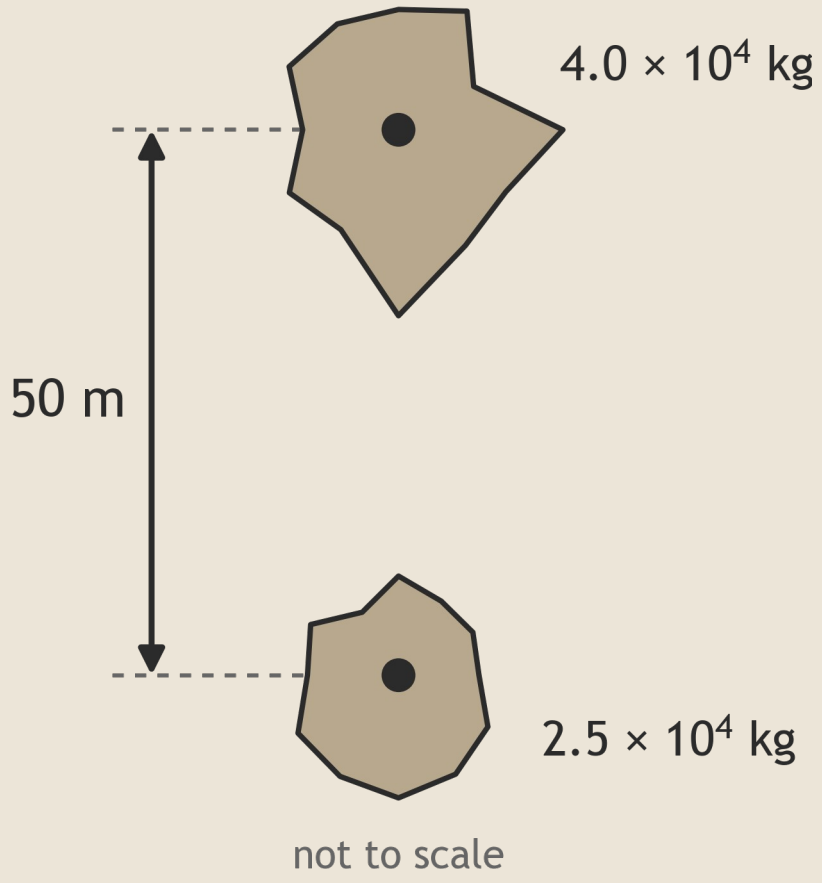
r = R + h



Worked Example 1

Two asteroids have masses of 4.0×10^4 kg and 2.5×10^4 kg. Their centres are 50 m apart.

Calculate the gravitational force between the asteroids. **3**



Worked Example 1: Marking

Question	Expected response	Max mark	Additional guidance
1	$F = Gm_1m_2/r^2$ (1 mark) $F = 6.67 \times 10^{-11} \times 4.0 \times 10^4 \times 2.5 \times 10^4 / 50^2$ (1 mark) $F = 2.7 \times 10^{-5}$ N (1 mark)	3	Accept: 3×10^{-5}, 2.67×10^{-5}, 2.668×10^{-5}

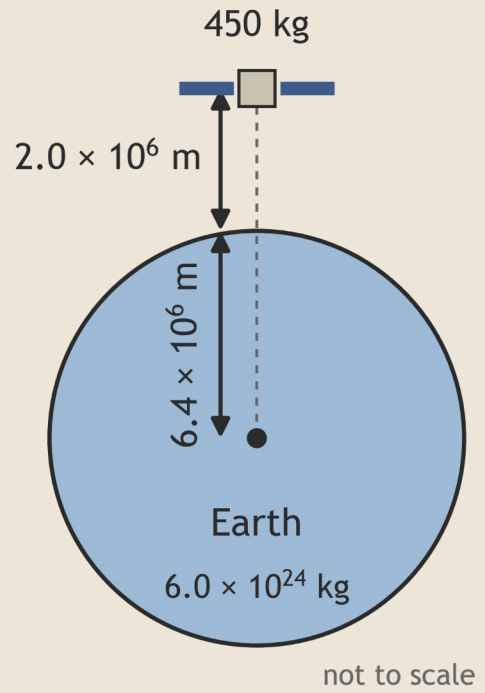
Worked Example 2

A satellite of mass 450 kg orbits 2.0×10^6 m above the surface of the Earth.

The mass of the Earth is 6.0×10^{24} kg and the radius of the Earth is 6.4×10^6 m.

(a) **Determine** the distance between the centre of the Earth and the satellite. **1**

(b) **Calculate** the gravitational force between the Earth and the satellite. **3**



not to scale

Worked Example 2: Marking

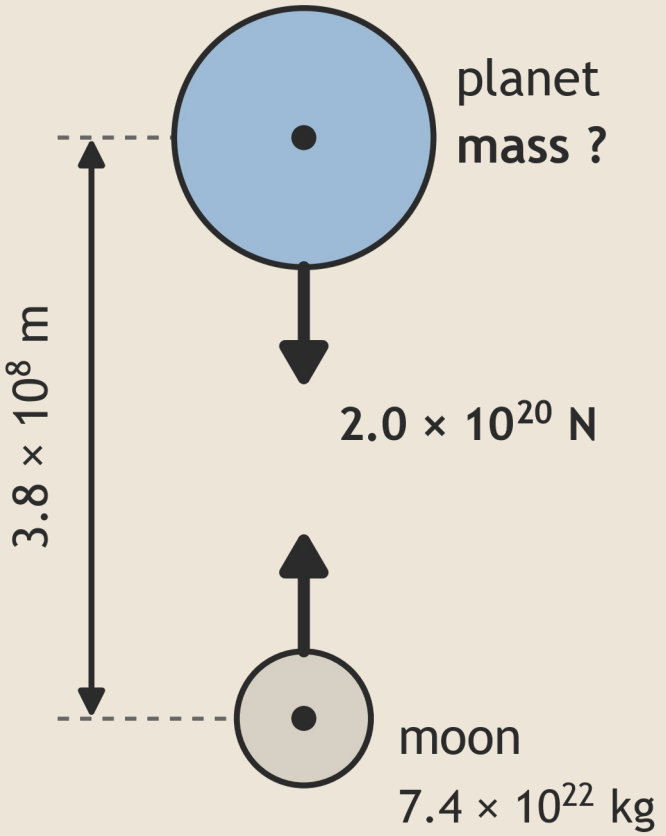
Question	Expected response	Max mark	Additional guidance
(a)	$(r = 6.4 \times 10^6 + 2.0 \times 10^6)$ $= 8.4 \times 10^6$ m (1 mark)	1	
(b)	$F = Gm_1m_2/r^2$ (1 mark) $F = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 450 / (8.4 \times 10^6)^2$ (1 mark) $F = 2.6 \times 10^3$ N (1 mark)	3	Or consistent with (a) Accept: 3×10^3, 2.55×10^3, 2.552×10^3

Worked Example 3

A moon of mass 7.4×10^{22} kg orbits a planet. The distance between their centres is 3.8×10^{8} m.

The gravitational force between them is 2.0×10^{20} N.

Calculate the mass of the planet. **3**



Worked Example 3: Marking

Question	Expected response	Max mark	Additional guidance
3	$F = Gm_1m_2/r^2$ (1 mark) $2.0 \times 10^{20} = 6.67 \times 10^{-11} \times m_1 \times 7.4 \times 10^{22} / (3.8 \times 10^8)^2$ (1 mark) $m_1 = 5.9 \times 10^{24}$ kg (1 mark)	3	Accept: 6×10^{24}, 5.85×10^{24}, 5.851×10^{24}

Your Turn



1. **Calculate** the force between lead spheres of 160 kg and 0.73 kg, 0.23 m apart.

   1.5×10^{-7} N

2. A 1200 kg satellite orbits 6.0×10^{5} m above the Earth. **Calculate** the force on it.

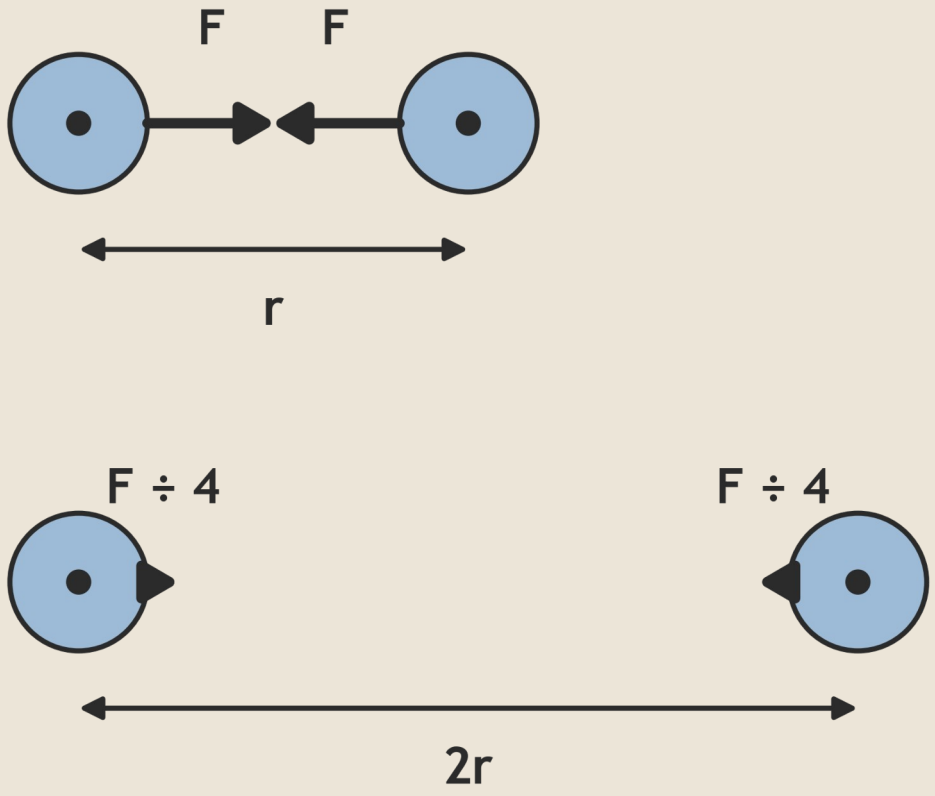
   r = 7.0×10^{6} m;
   9.8×10^{3} N

3. The Sun pulls the Earth (6.0×10^{24} kg) with 3.5×10^{22} N from 1.5×10^{11} m. **Calculate** the Sun's mass.

   2.0×10^{30} kg

Inverse Square: Change m or r

Change	Force
Double one mass	F × 2
Double both masses	F × 4
Double r	F ÷ 4
Halve r	F × 4



g at a Satellite

Weight

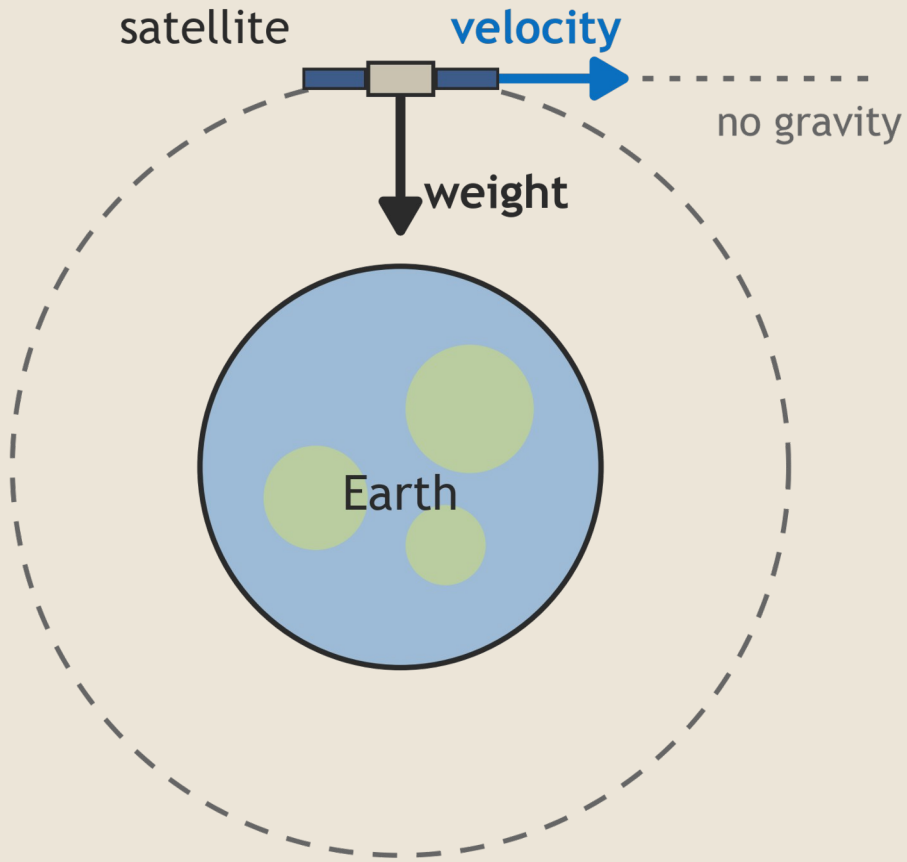
is the **gravitational force** on a mass

So

W = mg at any height

g at a satellite

g = F ÷ m, less than 9.8 N kg^{-1}



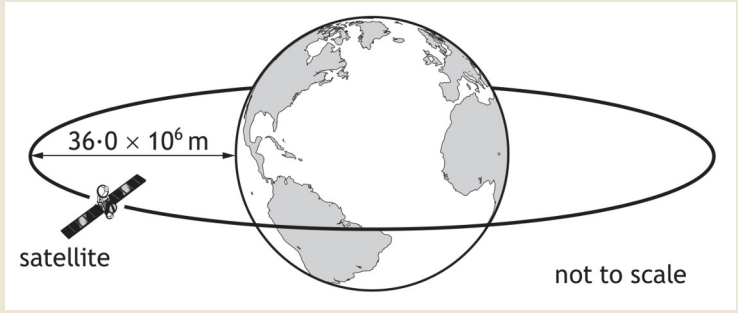
Past Paper: 2019 Paper 2 Q4(a)(b)

4. A communications satellite orbits the Earth at a height of 36.0×10^6 m above the surface of the Earth.

   The mass of the Earth is 6.0×10^{24} kg and the radius of the Earth is 6.4×10^6 m.

   (a) Determine the distance between the centre of the Earth and the satellite. **1**

   (b) The gravitational force of attraction between the Earth and the satellite is 57 N.
   Calculate the mass of the satellite. **3**



Scan: page 12.

Marking: 2019 Paper 2 Q4(a)(b)

Question	Expected response	Max mark	Additional guidance
4 (a)	(distance = $6.4 \times 10^6 + 36.0 \times 10^6$) $= 42.4 \times 10^6$ m (1 mark)	1	
(b)	$F = Gm_1m_2/r^2$ (1 mark) $57 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times m_2 / (42.4 \times 10^6)^2$ (1 mark) $m_2 = 260$ kg (1 mark)	3	Or consistent with (a) Accept: 300, 256, 256.1

Past Paper: 2019 Paper 2 Q4(c)(d)

4. (continued)

(c) Determine the value of the Earth's gravitational field strength g at the satellite. **3**

(d) A second satellite has a **quarter** of the mass of the first satellite.
The distance from the centre of the Earth to the second satellite is **half** the distance from the centre of the Earth to the first satellite.
State how the gravitational force of attraction between the second satellite and the Earth compares to the gravitational force of attraction between the first satellite and the Earth.
Justify your answer. **3**

Marking: 2019 Paper 2 Q4(c)

Question	Expected response	Max mark	Additional guidance
4 (c)	W = mg (1 mark) 57 = 260 × g (1 mark) g = 0.22 N kg^{-1} (1 mark)	3	Or consistent with (b) Accept: 0.2, 0.219, 0.2192 Accept: F = mg Do not accept: F = ma - (0 marks)

Marking: 2019 Paper 2 Q4(d)

Question	Expected response	Max mark	Additional guidance
4 (d)	Force is the same (1 mark) ¼ the mass has an effect of quartering the force (1 mark) ½ the orbital height has an effect of quadrupling the force (1 mark)	3	Look for this statement first - if incorrect or missing then 0 marks. Can justify by calculation Correct substitution of ½r and ¼m or consistent with (a) and (b) (1) Correct final answer (1)

Past Paper: 2023 Paper 1 Q6

A satellite of mass 620 kg is placed into an Earth orbit of radius 23 000 km.
The mass of the Earth is 6.0×10^{24} kg.
The gravitational force that the satellite experiences from the Earth in this orbit is

A 4.7×10^{2} N

B 4.7×10^{8} N

C 1.1×10^{10} N

D 1.1×10^{13} N

E 6.9×10^{13} N

Answer A: r in metres, 2.3×10^{7} m, then square it.

Scan: page 6.

Past Paper: 2018 Q5

Enceladus is a moon of Saturn. The mass of Enceladus is 1.08×10^{20} kg.
The mass of Saturn is 5.68×10^{26} kg.
The gravitational force of attraction between Enceladus and Saturn is 7.24×10^{19} N.
The orbital radius of Enceladus around Saturn is

A 2.38×10^{8} m
B 9.11×10^{13} m
C 5.65×10^{16} m
D 8.30×10^{27} m
E 3.19×10^{33} m

Answer A: $r^2 = Gm_1m_2/F$, then the square root.

Scan: page 4.

Past Paper: 2016 Q5

A planet orbits a star at a distance of 3.0×10^{9} m.
The star exerts a gravitational force of 1.6×10^{27} N on the planet.
The mass of the star is 6.0×10^{30} kg.
The mass of the planet is

A 2.4×10^{14} kg
B 1.2×10^{16} kg
C 3.6×10^{25} kg
D 1.6×10^{26} kg
E 2.4×10^{37} kg

Answer C: $m = Fr^2/(GM)$: square r first.

Scan: page 5.

What Candidates Get Wrong

Substituting

square r; use G, not g (2024 Q4)

Statement first

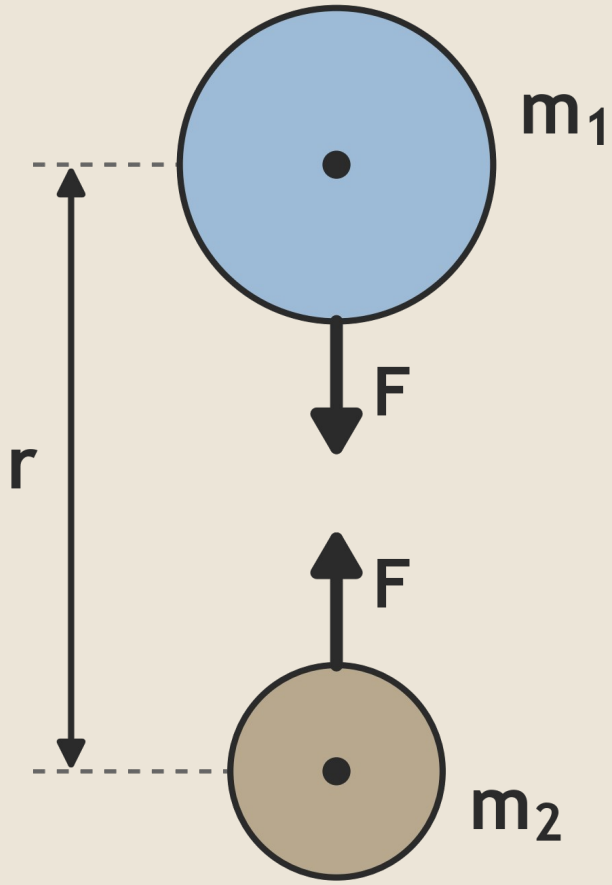
or 0 marks (2019 Q4(d), 38 %)

g at a satellite

W = mg; F = ma scores 0 (2019 MI)

Satellite force

many could not find it (2023 Q6)



r: centre to centre

Look back — by the end of this part...



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Course specification — 1 of 1

- Use of Newton's Law of Universal Gravitation to solve problems involving force, masses and their separation.

Scan: the course specification, page 8.

Credits (1 of 2)

Title: the ISS: NASA

Past Paper: 2019 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2019 Higher Physics Paper 2, Q4

Marking: 2019 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2019 Higher Physics Paper 2, Q4

Past Paper: 2019 Paper 2 Q4(c)(d): © Qualifications Scotland (SQA). 2019 Higher Physics Paper 2, Q4

Marking: 2019 Paper 2 Q4(c): © Qualifications Scotland (SQA). 2019 Higher Physics Paper 2, Q4

Marking: 2019 Paper 2 Q4(d): © Qualifications Scotland (SQA). 2019 Higher Physics Paper 2, Q4

2023 Paper 1 Q6: © Qualifications Scotland (SQA). 2023 Higher Physics Paper 1, Q6

2018 Q5: © Qualifications Scotland (SQA). 2018 Higher Physics Section 1, Q5

2016 Q5: © Qualifications Scotland (SQA). 2016 Higher Physics Section 1, Q5

Credits (2 of 2)

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8

R Stewart - November 2026